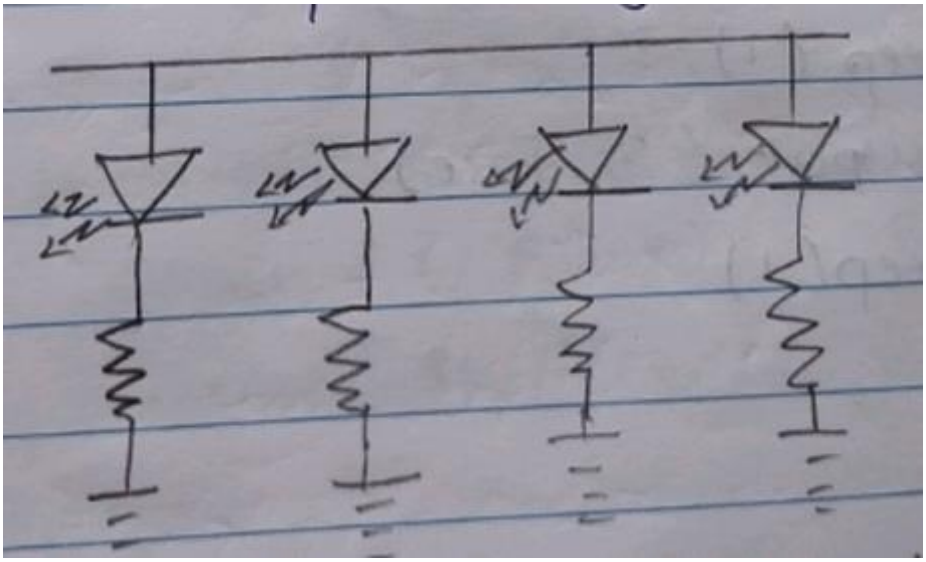
Practical no. 1

Aim – Create different blinking patterns with LEDs by connecting to GPIO pins of Raspberry Pi.

Apparatus – Breadboard, LED, Resister, Raspberry Pi 4, Jumper wires.

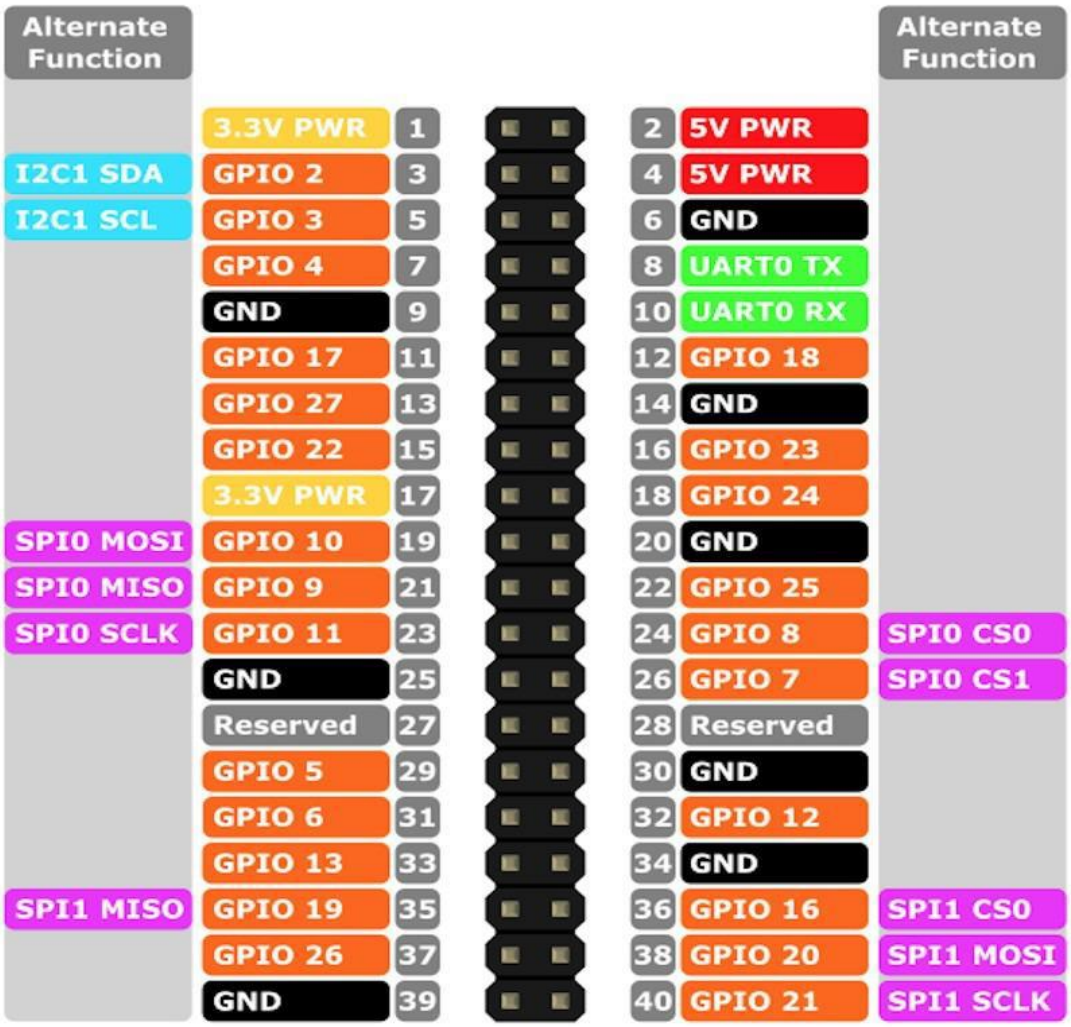
Theory -
Raspberry Pi is a small, low-cost, single-board computer developed by the Raspberry Pi Foundation (UK). It works just like a normal computer but is much smaller. Raspberry is just the project name. Pi comes from Python Programming Language, because Raspberry Pi was mainly designed to teach Python.
GPIO means - General Purpose Input Output
These are special pins that allow Raspberry Pi to communicate with electronic components.

Circuit diagram –



GPIO7 GPIO11 GPIO 13 GPIO15

Raspberry Pi 4 pin diagram -



Source Code –

```
import RPi.GPIO as x
import time
x.setwarnings(False)
x.setmode(x.BOARD)
x.setup(7, x.OUT)
x.setup(11, x.OUT)
x.setup(13, x.OUT)
x.setup(15, x.OUT)
while True:
for i in range(3):
        x.output((7,11,13,15), (1,1,1,1))
        time.sleep(0.5)
        x.output((7,11,13,15), (0,0,0,0))
        time.sleep(0.5)
for i in range(3):
        x.output((7,11,13,15), (1,1,0,0))
        time.sleep(0.5)
```

```
        x.output((7,11,13,15), (0,0,1,1))
        time.sleep(0.5)
for i in range(3):
        x.output((7,11,13,15), (1,0,0,0))
        time.sleep(0.5)
        x.output((7,11,13,15), (1,1,0,0))
        time.sleep(0.5)
        x.output((7,11,13,15), (1,1,1,0))
        time.sleep(0.5)
        x.output((7,11,13,15), (1,1,1,1))
        time.sleep(0.5)
        x.output((7,11,13,15), (1,1,1,0))
        time.sleep(0.5)
        x.output((7,11,13,15), (1,1,0,0))
        time.sleep(0.5)
        x.output((7,11,13,15), (1,0,0,0))
        time.sleep(0.5)
        x.output((7,11,13,15), (0,0,0,0))
        time.sleep(0.5)
```

Output –

